

Optimal Economical Schedule of Hydrogen-Based Microgrids With Hybrid Storage Using Model Predictive Control

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Abstract—The electricity market rules determine the energy prices in the day-ahead market, matching offers from generators to bids from consumers. The unpredictability of renewable energy combined with the penalty deviations used in the regulation market makes it difficult for clean energy to play an important role in the electricity market. The high density of hydrogen as an energy storage system (ESS) appears to be one solution to the problems outlined. There is still not a perfect ESS, everyone has different limitations from the point of view of time autonomy, time response, degradation issues, or acquisition cost. The design of a hybrid energy storage management system emerges as a technological solution to the problems commented. The development of an optimal control for renewable energy microgrids with hybrid ESS is carried out using model predictive control (MPC). The MPC techniques allow maximizing the economical benefit of the microgrid, minimizing the degradation causes of each storage system, and fulfilling the different system constraints. In order to capture both continuous/discrete dynamics and switching between different operating conditions, the plant is modeled with the framework of mixed logic dynamic. The MPC problem is solved within mixed-integer quadratic programming.

Index Terms—Energy management, energy storage, hydrogen.

NOMENCLATURE

C	Capacity (Wh).
CC	Capital cost (€).
Cost	Hourly economic cost (€/h).
Cycles	Number of life cycles.
h_i	Hour i .
Hours	Number of life hours.

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LOH	Level of hydrogen (Nm^3).
m_j	Minute j .
SOC	State of charge [per unit (p.u)].
SH	Schedule horizon.
t_k	Generic instant.
P	Electric power (W).
T_s	Sample period.
z	Electric power formulated as mixed logic dynamic (MLD) variable (W).
δ	ON-/OFF-state.
η	Performance (p.u).
χ	Logical degradation state.
ϑ	MLD power variation in degradation state (W).
Sel_z	Electrolyzer consumption (Nm^3/Wh).
Sfc	Fuel cell consumption (Wh/Nm^3).
σ	Logical variable start-up/shutdown state.
Γ	Energy price(€).
Subscripts	
bat	Battery.
ch	Charge.
degr	Degradation.
dev	Deviation.
dis	Discharge.
down	Down.
elz	Electrolyzer.
fc	Fuel cell.
grid	Main grid.
h_2	Hydrogen.
load	Load.
pur	Purchase of energy.
pv	Photovoltaic (PV) system.
reg	Regulation.
rem	Remaining power.
sale	Sale of energy.
uc	Ultracapacitor.
up	Up.
wt	Wind turbine.
μgrid	Microgrid.
Superscripts	
DM	Daily market.
IM	Intraday market.
MM	Deviation management market.
RM	Regulation service market.
max	Maximum value.
min	Minimum value.

on	ON-state.
off	OFF-state.
ref	Reference.
sch	Schedule.
shutdown	Shutdown.
startup	Start-up.

I. INTRODUCTION

THE electricity market is a complex process where the electricity production system provides a total amount of electricity, in each instant, corresponding to a varying load from the electricity consumption. Electricity exchanges first take place in the day-ahead (spot) market, where participants¹ (sellers and buyers) have to propose, before gate closure, their quantity–price bids over the following delivery period. Participants are then financially responsible for any deviation from the contract. Certain electricity pools also integrate intraday markets, where it is possible to take corrective actions. The regulation market, which is managed by the system operator,² ensures the real-time balance between generation and load. For the fast load variations, as well as unforeseen problems with production capacity, there are reserves at the system operator's disposal [1], [2].

The introduction of energy storage system (ESS) in the paradigm of smart grids introduces new possibilities in the electrical market. The high energy density of hydrogen as an energy carrier will play an important role; however, the robust performance under transient conditions and slow response are the main barriers for its technological implantation. Batteries and supercapacitors have a better transient response; however, their low energy density does not give enough autonomy to the system. Every ESS has its own constraints, but it is also subject to a functional cost due to different aspects such as capital cost, operation, and maintenance cost or degradation issues [3]–[5].

Hydrogen storage systems are composed of an electrolyzer, a fuel cell, and a storage tank. Both electrolyzers and fuel cells have different degradation mechanisms. Fluctuations in the current applied to electrolyzers produce the following: 1) variations in the differential pressure, producing reverse conduction of hydrogen to the oxygen separator or *vice versa*; 2) small concentrations of hydrogen in oxygen or oxygen in hydrogen (> 4%), which can produce highly explosive reactions; 3) losses in the mechanical wear of the membrane; and 4) chemical degradation of the membrane via radical attack. On/Off cycling of electrolyzers produces the following: 1) chemical and mechanical degradation due to the temperature and pressure cycling; 2) chemical degradation caused by uncontrolled stack polarity; and 3) production of hydrogen peroxide at cathode after turn off with following membrane and carbon

carrier oxidation [6]–[9]. Meanwhile, fluctuant load cycling of fuel cells produces the following: 1) catalyzer (platinum) particles dissolving in the cathode due to the potential cycling; 2) mechanical wear of the membrane due to the thermal and humidity cycling; 3) chemical degradation of the membrane via radical attack due to the time at open-circuit voltage; 4) starvation phenomena when the current drawn from a fuel cell is not in the correct proportion to the reacted oxygen; and 5) losses in the correct humidity in the membrane. Start/Stop cycling produces the following: 1) carbon corrosion in the electrodes; and 2) deactivation of the catalysis layer [10]–[12].

Batteries overcharging produces the formation of permanent oxides with the loss of electrode active material. High depth of discharge can damage the life cycle of the battery. High charging and discharging current produces a specific degradation mechanism in batteries increasing the resistance related to the electrode–electrolyte [12]–[14].

Ultracapacitors can be charged substantially faster than conventional batteries and with a higher number of life cycles. However, ultracapacitors have a low energy density and behave like a short circuit when exposed to low levels of SOC. High states of charge can also damage this technology [15].

Benefits of hybrid ESS as a combination of energy, power density, cost, and life cycle have been manifested in several studies such as [12] and [16]–[21]. The existing literature exposes how the use of hybrid ESS can satisfy a major range of applications, improving the lifetime of the single components. The optimal use of the hybrid ESS requires the development of a controller, which takes into account all the constraints, limitations, degradation issues, and the economical cost of each ESS. The economical costs of the use of each storage system are introduced by Geer *et al.* [22], where an evaluation of the economics of operating a wind/hydrogen bus system is exposed. Dufo-López *et al.* [23] considered a strategy applied to a stand-alone microgrid composed of renewable sources [wind, PV, and hydro], batteries, a fuel cell, an ac generator, and an electrolyzer, which optimizes the control of the hybrid system, minimizing the total cost throughout its lifetime using genetic algorithms. Stevens [12] quantified the degradation cost of a Li-ion battery and a fuel cell used in a hybrid electrical vehicle. Bergen [24] focused his research on the interest of the degradation issues and life cycle of electrolyzers solving the control algorithm using heuristic algorithms. Heuristic algorithms applied to the economical dispatch of microgrids are presented by Chakraborty and Simoes [25], who solved the economical dispatch of a PV microgrid with batteries only, such as ESS. In [26]–[28], a dead-band hysteresis algorithm is used to decide the use of the battery ESS or the hydrogen ESS. Gu *et al.* [29] made an optimization control algorithm for the combined heat and power of a microgrid with batteries and a fuel cell fed with natural gas, neglecting degradation aspects of the ESS. In [30] and [31], different strategies in the control of microgrids with hybrid storage using heuristical methods are presented. Microgrid optimal scheduling considering multiperiod islanding constraints is solved using mixed-integer programming in the study carried out by Khodaei [32]. A self-organizing computing framework, which is based on self-organizing agents, for solving the fundamental control

¹Market participants are undertakings that are authorized to act directly in the electric power market as buyers and sellers of electricity.

²The role of the system operator in a wholesale electricity market is to manage the security of the power system in real time and coordinate the supply of and demand for electricity, in a manner that avoids fluctuations in frequency or interruptions of supply. On the other hand, the market operator is in charge of the wholesale transactions (bids and offers) in the day-ahead and intraday markets.

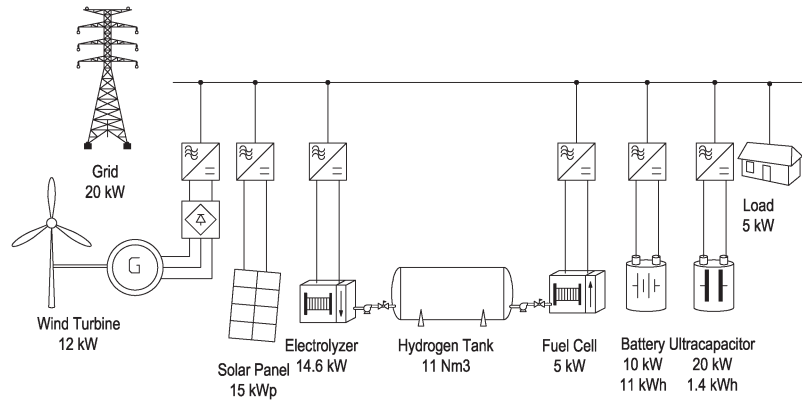


Fig. 1. Microgrid object of this study.

and monitoring problems of a microgrid, is presented in [33]. The control problem of coordinated energy management of networked microgrids is treated in [34]. Jiang and Fei [35] proposed an energy ecosystem facilitated by microgrid management with hierarchical agents and optimization where updated external information and user's task preference are considered to make the control decisions. Nguyen and Le [36] proposed a framework for optimizing energy trading operations of a microgrid aggregator in the energy market.

In the paper presented by Ferrari-Trecate *et al.* [37], the development of model predictive control (MPC) for hybrid cogeneration power plants is carried out introducing the MLD framework. MPC applied to distributed energy resources with battery ESS is developed by Negenborn *et al.* [38], introducing an MLD model but without taking into account other kinds of ESS or the degradation issues associated with the battery. Korpås [39] used MPC to integrate hydrogen ESS in the electrical market not considering other ESS possibilities, neglecting degradation issues associated with the ESS. An MPC controller is developed by Trifkovic *et al.* for the use of fuel cell and electrolyzer in [40]. Del Real *et al.* [41] and Valverde *et al.* [42] developed in their studies MPC strategies applied to the field of microgrid optimization without considering the degradation issues of the ESS. Mixed-integer linear programming (MILP), including operational costs in the ESS, is used in [43]–[46] without integrating the degradation issues in the ESS. Vahidi and Greenwell, in their studies [47], [48], applied the MPC to control the load sharing of a hybrid ESS composed of a fuel cell and an ultracapacitor, including some degradation issues, but these studies do not include the connection to the grid or the start-up/shutdown degradation issues associated with the fuel cell. There are almost no studies where the application of the trihybrid ESS ultracapacitor–battery–hydrogen in the field of the smart grid is considered, although this configuration is widely studied in the field of electrical vehicles [16]–[21].

In this paper, the optimization problem of the energy inside a renewable energy microgrid with different kinds of ESSs, which exchanges energy with the main grid, is developed and solved with the use of MPC techniques. The controller includes the operational cost, degradation issues, and constraints associated to the hybrid ESS, resulting in the main contribution of this paper. All the schedule steps of the market operator and

the system operator are considered in comparison with previous work, which have only solved one of the steps of the electricity market.

This paper is organized as follows. Section II explains the experimental setup of this study. Section III presents the controller design. Section IV presents the results of the energetic schedule carried out by the MPC controller in the microgrid. Section V makes a comparison with previous methods used in the field of microgrid economic dispatch. Finally, Section VI outlines the conclusions.

II. EXPERIMENTAL SETUP

In this paper, the control strategy for a grid-connected microgrid (with the possibility to work as an isolated microgrid) integrating wind turbine power, PV generation, and hybrid ESS composed of ultracapacitors, batteries, a fuel cell, and an electrolyzer is presented. The microgrid is described in Fig. 1 and Table I.

The renewable energy data have been provided by the meteorological station, from the company Geonica, model MTD 3008, located at the installations of the Spanish National Center on Hydrogen, which has been collecting data every 10 min since 2009. The load profile corresponds to the collected data of a domestic home located in Puertollano (Ciudad Real, Spain), whose maximum contracted power with the electrical company Union Fenosa is 5 kW. The energy price data have been given by the Iberian Market Operator (OMIE). The exposed results correspond to the day of February 28, 2014.

III. CONTROLLER DESIGN

The different timescales of the electricity markets make a whole control algorithm necessary solving the long-term horizon schedule made by the daily market (38 h ahead) and the different sessions of the intraday market (28–29 h), the deviation management market (5 h), and the regulation service market (3 h) to the real-time load sharing (seconds) (more information about the electrical market can be found in [52] and [53]). The horizon schedule also determines the most appropriate ESS technology to be used. The cost and constraints of every ESS based on the life and load cycling degradation of each ESS, as

TABLE I
MICROGRID COMPONENTS

Solar Panel (ISOFOTON, Model: ISOFOTON-150)
Serial Modules: 38, Parallel Modules: 3
$P = 15 \text{ kWp}$, $V_{\text{module,max}} = 18.5 \text{ V}$, $I_{\text{module,max}} = 8.12 \text{ A}$
Wind Turbine (BORNAY, Model: BORNAY BK-12)
Rotor Diameter= 7 m, Power=12 kW, Nominal Wind Speed=12 m/s
PEM Electrolyzer (PROTON, Model: H2m)
Net Production Rate: 2 Nm ³ /hr, Pressure=1.5 to 4 barg,
$\varsigma = 7.3 \text{ kWh/Nm}^3$, CC= 8.22 €/kW, $\text{Cost}_{o\&m,elz} = 2 \text{ m€/h}$
$\text{Cost}_{\text{startup},elz} = 0.123 \text{ €}$, $\text{Cost}_{\text{shutdown},elz} = 0.0062 \text{ €}$
$\text{Cost}_{\text{degr},elz} = 0.05 \text{ €/W}$, Lifetime= 10000 hours
PEM Fuel Cell (NEDSTACK, Model: FCS 5-HP)
$P = 5 \text{ kW}$, Pressure=0.15 to 0.30 barg
$\varsigma = 0.768 \text{ Nm}^3/\text{kWh}$, CC=30 €/kW, $\text{Cost}_{o\&m,fc} = 1 \text{ m€/h}$
$\text{Cost}_{\text{startup},fc} = 0.01 \text{ €}$, $\text{Cost}_{\text{shutdown},fc} = 0.005 \text{ m€/h}$,
$\text{Cost}_{\text{degr},fc} = 0.01 \text{ €/W}$, Lifetime= 10000 hours
Hydrogen Tank (IDESIA)
Volume = 11 Nm ³ , Pressure = 30 bar
Li-Ion Battery (SAFTBATTERIES, Model: Synerion)
Serial Modules: 5, $C_{\text{mod}} C/5$ at +25°C = 43.5 Ah, $C = 11 \text{ kWh}$
$P = 10 \text{ kW}$, $V = 210$ to 282 V
$\eta_{ch} = 0.90$, $\eta_{dis} = 0.95$, CC=125 €/kWh, Life cycles=3000,
$\text{Cost}_{\text{degr},dis} = 10^{-9} \text{ €/W}^2\text{h}$, $\text{Cost}_{\text{degr},ch} = 10^{-9} \text{ €/W}^2\text{h}$
Ultracapacitor(MAXWELL, Model: BMOD0063 P125 B08)
Serial Modules: 10, $C_{\text{mod}} = 68 \text{ F}$, $V_{\text{mod}} = 125 \text{ V}$,
$P = 20 \text{ kW}$, $\eta_{ch} = 0.97$, $\eta_{dis} = 0.99$, $C_{\text{total}} = 1475 \text{ Wh}$

Data based on U.S. Department of Energy (DOE) final target [49], [50], [51].

well as the deviation schedule penalty with respect to the main grid, are considered using a four-level-cascade MPC controller. The block diagram of the different levels of the controller is detailed in Fig. 2. The controller was implemented in MATLAB using the solver CPLEX.

The horizon of each control level is dictated by the timescale of the corresponding market. In addition to the CCs shown in Table I, the MPCs are tuned giving bigger weights to degradation costs of fuel cell and electrolyzer, since these devices are more sensitive to degradation. The next order in the weighting factors is the battery, and the last one is the ultracapacitor, which allows more fluctuant charge conditions.

A. Model of the Plant

The model of the plant has two parts: the remaining energy and price forecast model and the evolution of the state variables of the microgrid (LOH, SOC_{uc}, SOC_{bat}).

1) Forecast of the Remaining Energy in the Microgrid and Price Evolution: The model is based on an artificial neural network (ANN) developed with the autoregressive integrated moving average model (ARIMA) using the time series method. Its result is the forecasting of the wind power generation, the solar power generation, and the load consumption, using as input the meteorological variables: solar irradiance G , ambient temperature T , atmospheric pressure P , relative humidity H , and wind speed, including the wind direction W . The ANN is a three-layer model (input layer, hidden layer, and output layer). The input layer neurons are the profile of the last hour for the indicated measures. The first step of the development is to identify the most similar profile for each neuron at exactly the same time that the prediction is made over the previous two weeks. The next step is to compare, for the same day, the

most similar profile for each variable in all the following days during a month for the same instant in the last two years. With the selected data, the hidden layer simply selects those profiles that are similar to a maximum error value of 15%. With these selected data, the model takes the irradiance, the temperature, and the wind speed in the required prediction horizon for each neuron. Finally, the average of the prediction made is taken for the selected neurons in the hidden layer. The output layer prediction is an average value where the recent year variables and the temperatures are more weighted than the other neurons. In the case that no neuron profile is similar to the present profile with an error of below 15%, the most similar temperatures are taken for the prediction. The output of the model gives the prediction of the irradiance, the temperature, and the wind speed for the horizon prediction. With these data, the power production for the PV array P_{pv} and wind turbine generator P_{wt} can be easily predicted following the models explained by Villalva and Gazoli [55] and Iov *et al.* [56]. The price forecast and the load profile P_{load} are done by taking a statistical average of the last month using four different profiles. The first one corresponds to the working days. The second profile corresponds to Friday after noon. The third profile corresponds to Saturday, and the forth profile corresponds to Sunday. The result of the model gives the prediction for the remaining power and the price energy forecast for each instant of discretization of the prediction horizon for every level of control of the microgrid, i.e.,

$$P_{\text{rem}}(t_k) = P_{pv}(t_k) + P_{wt}(t_k) - P_{load}(t_k). \quad (1)$$

2) Linear State-Space Model of the Plant: The dynamic of the state variables, which are the level of the storage systems, is given by

$$\begin{aligned} \text{SOC}_{\alpha}(t_{k+1}) = \text{SOC}_{\alpha}(t_k) &+ \frac{\eta_{ch,\alpha} \cdot P_{ch,\alpha}(t_k) \cdot T_s}{C_{\alpha}} \\ &+ \left. \frac{P_{dis,\alpha}(t_k) \cdot T_s}{\eta_{dis,\alpha} \cdot C_{\alpha}} \right|_{\alpha=\text{bat,uc}} \end{aligned} \quad (2)$$

$$\begin{aligned} \text{LOH}(t_{k+1}) = \text{LOH}(t_k) &+ \frac{P_{elz}(t_k) \cdot \delta_{elz}(t_k) \cdot T_s}{S_{elz}} \\ &- \varsigma_{fc} \cdot P_{fc}(t_k) \cdot \delta_{fc}(t_k) \cdot T_s. \end{aligned} \quad (3)$$

B. Daily Market Microgrid MPC Controller

The purpose of the daily market is to handle electricity transactions for the following day through the presentation of electricity sale and purchase bids by market participants. Bids made by these sellers are presented to the market operator and will be included in a matching procedure that will affect the daily programming schedule corresponding to the following day. The market operator matches electricity power purchase and sale bids (received before 10 A.M. on each day). As shown in Fig. 2, the outputs of the controllers are the reference power values for all the ESS, for each hour of the day. The ultracapacitor ESS does not have enough energy density to schedule long-term control horizon; thus, it is not considered. Therefore,

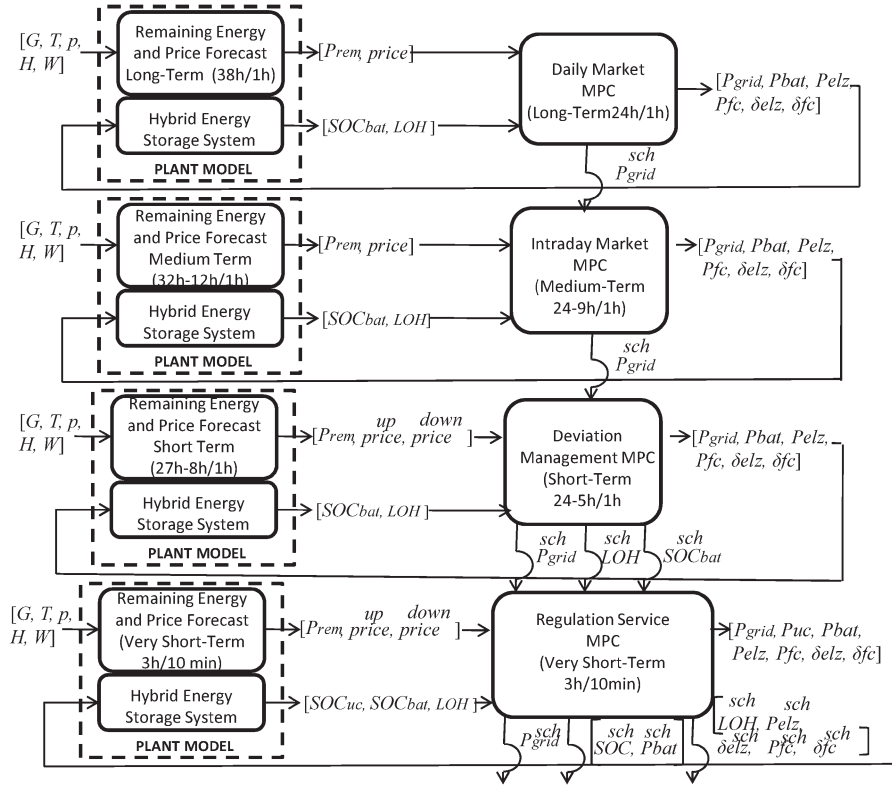


Fig. 2. MPC controller block diagram.

the following cost function is minimized (using $T_s = 1$ h as sample period):

$$J = \sum_{h_i=1}^{24} (J_{\text{grid}}(h_i) + J_{\text{bat}}(h_i) + J_{\text{elz}}(h_i) + J_{\text{fc}}(h_i)). \quad (4)$$

1) Grid Cost Function: The cost function of the grid is given by the economical revenue of selling energy to the grid or by the economical cost of buying energy from the grid, i.e.,

$$J_{\text{grid}} = \sum_{h_i=1}^{24} (-\Gamma_{\text{sale}}^{\text{DM}}(h_i) \cdot P_{\text{sale}}(h_i) + \Gamma_{\text{pur}}^{\text{DM}}(h_i) \cdot P_{\text{pur}}(h_i)) \cdot T_s \quad (5)$$

where the sale and purchase of energy with the grid are expressed by the following equations, replacing subscripts α , β , and γ with $\alpha = \text{sale}$, $\beta = \text{pur}$, and $\gamma = \text{grid}$:

$$P_{\alpha}(t_k) = \begin{cases} P_{\gamma}(t_k), & P_{\gamma}(t_k) \geq 0 \\ 0, & P_{\gamma}(t_k) < 0 \end{cases} \quad (6)$$

$$P_{\beta}(t_k) = \begin{cases} 0, & P_{\gamma}(t_k) > 0 \\ P_{\gamma}(t_k), & P_{\gamma}(t_k) \leq 0. \end{cases} \quad (7)$$

These piecewise functions are introduced in the MPC controller using the transformation explained by Bemporad and Morari [54], resulting in the MLD constraints expressed in the inequalities (36)–(40). All the constraints of the controller are exposed in Section III-F.

2) Batteries Cost Function: The manufacturers of the batteries quantify the life of this ESS as a function of the number of the charge and discharge cycles. In order to take into account the degradation issues associated with batteries, Arora *et al.* in the study [14] modeled the batteries degradation like a function of a film growing, i.e., ι , which increases the internal resistance of the lithium-ion battery, as expressed by

$$\frac{d\iota}{dt} = \kappa \cdot I_{\text{bat}}. \quad (8)$$

This depends on the battery current and a series of constant parameters, which depend on the battery [In (8), κ is a parameter that depends on the manufacturer's design].

In order to take this fact into account, a second term is introduced in the cost function associated to the battery, where the cost does not only depend on the cycle used but also quantifies this film growing, i.e.,

$$J_{\text{bat}} = \sum_{h_i=1}^{24} \left(\frac{\text{CC}_{\text{bat}}}{2 \cdot \text{Cycles}_{\text{bat}}} P_{\text{bat, ch}}(h_i) \cdot T_s \cdot \eta_{\text{bat, ch}} + \text{Cost}_{\text{degr, ch}} \cdot P_{\text{bat, ch}}^2(h_i) + \frac{\text{CC}_{\text{bat}}}{2 \cdot \text{Cycles}_{\text{bat}}} \frac{P_{\text{bat, dis}}(h_i) \cdot T_s}{\eta_{\text{dis, bat}}} + \text{Cost}_{\text{degr, dis}} \cdot P_{\text{bat, dis}}^2(h_i) \right). \quad (9)$$

The charge and discharge of the batteries can be expressed with equations (6) and (7), replacing subscripts α , β , and γ with

$\alpha = \text{ch}$, $\beta = \text{dis}$, and $\gamma = \text{bat}$. Similar assumptions can be done with the ultracapacitor. These expressions result in the constraints expressed in (41)–(45).

3) Hydrogen Cost Function: Manufacturers of electrolyzers and fuel cells give the lifetime expression of these kinds of systems as a function of the number of life hours. Start-up and shutdown cycles and fluctuating load conditions can seriously affect these devices. In the same way, in the case of the batteries, the operation and maintenance cost is minimum, but in this kind of ESS, it must be considered. Thus,

$$J_{\text{elz}}(h_i) = \left(\frac{\text{CC}_{\text{elz}}}{\text{Hours}_{\text{elz}}} + \text{Cost}_{o\&m,\text{elz}} \right) \delta_{\text{elz}}(h_i) + \text{Cost}_{\text{startup},\text{elz}} \cdot \sigma_{\text{elz}}^{\text{on}}(h_i) + \text{Cost}_{\text{shutdown},\text{elz}} \cdot \sigma_{\text{elz}}^{\text{off}}(h_i) + \text{Cost}_{\text{degr},\text{elz}} \cdot \vartheta_{\text{elz}}^2(h_i) \quad (10)$$

$$J_{\text{fc}}(h_i) = \left(\frac{\text{CC}_{\text{fc}}}{\text{Hours}_{\text{fc}}} + \text{Cost}_{o\&m,\text{fc}} \right) \delta_{\text{fc}}(h_i) + \text{Cost}_{\text{startup},\text{fc}} \cdot \sigma_{\text{fc}}^{\text{on}}(h_i) + \text{Cost}_{\text{shutdown},\text{fc}} \cdot \sigma_{\text{fc}}^{\text{off}}(h_i) + \text{Cost}_{\text{degr},\text{fc}} \cdot \vartheta_{\text{fc}}^2(h_i). \quad (11)$$

In equations (10) and (11), a series of auxiliary logical variables have been introduced. The start-up, i.e., $\sigma_j^{\text{on}}(t_k)$, and shutdown, i.e., $\sigma_j^{\text{off}}(t_k)$, states for the electrolyzer and the fuel cell are defined by

$$\sigma_j^{\text{on}}(t_k) = \max(\delta_j(t_k) - \delta_j(t_{k-1}), 0) |_{j=\text{elz},\text{fc}} \quad (12)$$

$$\sigma_j^{\text{off}}(t_k) = \max(\delta_j(t_{k-1}) - \delta_j(t_k), 0) |_{j=\text{elz},\text{fc}}. \quad (13)$$

Using the Karnaugh map rules, these equations can be expressed as logical relationships, i.e.,

$$\sigma_j^{\text{on}}(t_k) = \delta_j(t_k) \vee (\sim \delta_j(t_{k-1})) |_{j=\text{elz},\text{fc}} \quad (14)$$

$$\sigma_j^{\text{off}}(t_k) = \delta_j(t_{k-1}) \vee (\sim \delta_j(t_k)) |_{j=\text{elz},\text{fc}}. \quad (15)$$

Using the relationships defined by Bemporad and Morari [54], the next inequalities are introduced in the constraints of the MPC controller (47)–(52). The logical power variation $\vartheta_j(t_k)$ for the electrolyzer and the fuel cell is defined by the power variation in all the instants apart from those where the device is started up or shut down; this casuistic is defined by

$$\vartheta_j(t_k) = \Delta z(t_k) \cdot (\delta_j(t_k) \vee \delta_j(t_{k-1})) |_{j=\text{elz},\text{fc}} \quad (16)$$

where the logical power $z(t_k)$ of each device can be defined as the product of the power reference value and the logical ON/OFF-state resulting in the introduction of MLD constraints in the controller cited in the expressions (53)–(55), i.e.,

$$z_j(t_k) = P_j(t_k) \cdot \delta_j(t_k) |_{j=\text{elz},\text{fc}}. \quad (17)$$

A new auxiliary variable, which is called logic state of degradation by power variation, i.e., $\chi_j(t_k)$, is defined, resulting in the

constraints expressed in inequalities (56)–(58). The following equation gives, as result, the MLD constraints (59)–(61):

$$\chi_j(t_k) = (\delta_j(t_k) \vee \delta_j(t_{k-1})). \quad (18)$$

The constraint introduced in (46) avoids the fuel cell and the electrolyzer working at the same time.

C. Intraday Market MPC

The purpose of the intraday market is to respond, through the presentation of electricity power sale and purchase bids by market participants, to adjustments made to the final viable daily schedule. It is structured into different sessions.³ Market participants may only participate for the hourly periods corresponding to those included in the daily market in which they have participated [53]. In order to simplify the block diagram (see Fig. 2), only the first session has been included in the figure. In a similar way that the daily market outputs are considered as inputs for the intraday market MPC, the inputs of the rest of sessions correspond with the outputs of the previous session. The sample period used for this control level is $T_s = 1$ h. Thus,

$$J_{\text{grid}}(h_i) = -\Gamma_{\text{sale}}^{\text{DM}}(h_i) \cdot P_{\text{sale}}^{\text{sch}}(h_i) + \Gamma_{\text{pur}}^{\text{DM}}(h_i) \cdot P_{\text{pur}}^{\text{sch}}(h_i) + \Gamma_{\text{sale}}^{\text{IM}}(h_i) \cdot P_{\text{sale}}^{\text{IM}}(h_i) \cdot \delta_{\text{grid}}(h_i) - \Gamma_{\text{pur}}^{\text{IM}}(h_i) \cdot P_{\text{pur}}^{\text{IM}}(h_i) \cdot \delta_{\text{grid}}(h_i) \quad (19)$$

$$\delta_{\text{grid}}(t_k) = \begin{cases} 1, & P_{\text{grid}}^{\text{sch}}(t_k) \neq 0 \\ 0, & P_{\text{grid}}^{\text{sch}}(t_k) = 0. \end{cases} \quad (20)$$

Only the instants in which the energy exchange exists with the grid in the day-ahead market can be participated. This is done with the introduction of the logical variable $\delta_{\text{grid}}(t_k)$. The product of logical and dynamic variables resulting in the introduction of the auxiliary variables z_{sale} and z_{pur} with the constraints (53)–(55) is as follows:

$$P_{\text{grid}}(h_i) = P_{\text{sale}}^{\text{sch}}(h_i) - P_{\text{pur}}^{\text{sch}}(h_i) + z_{\text{sale}}(h_i) - z_{\text{pur}}(h_i). \quad (21)$$

Taking into account the considerations made for the intraday market level control of the microgrid, the next global cost function is introduced, whose SH is given by the prediction horizon of each intraday session. Cost functions defined for the batteries, the electrolyzer, and the fuel cell in the daily market MPC [see (9)–(11)] are also valid in the intraday market MPC. Thus,

$$J = \sum_{h_i=1}^{\text{SH}} (J_{\text{grid}}(h_i) + J_{\text{bat}}(h_i) + J_{\text{elz}}(h_i) + J_{\text{fc}}(h_i)). \quad (22)$$

³The intraday market is a complementary platform for the daily market, where electricity is traded to adjust the quantities traded in the daily market, consisting of a number of daily trading sessions. Each of the intraday market sessions forms a price for the hours, which is the object of negotiation in each session with an SH smaller than in the daily market.

D. Deviation Management Market MPC

Agents have to communicate the forecast of production deviations to the operator system (OS). If these deviations, together with variations with the wind power production forecast completed by the OS during the period between two intraday sessions, exceed 300 MW/h on average, it will result in the deviation management market. Offers from generators are requested on the opposite deviations provided in the system sense. Deviation management is remunerated with a competitive dual-price system, giving stronger incentives to deliver schedules [53]. The sample period used for this control level is $T_s = 1$ h. Thus,

$$J_{\text{grid}} = \sum_{h_i=t_o}^{\text{SH}} (\Gamma_{\text{up,dev}}^{\text{DM}}(h_i) \cdot z_{\text{up}}(h_i) + \Gamma_{\text{down,dev}}^{\text{DM}} \cdot z_{\text{down}}(h_i)) \quad (23)$$

where $z_j(h_i) = P_j(h_i) \cdot \delta_j(h_i) |_{j=\text{up,down}}$, being $\delta_j(h_i) |_{j=\text{up,down}}$ is active only in these instants when the deviation management market is executed by the OS. Similar cost functions defined for the batteries, the electrolyzer, and the fuel cell in the daily market MPC [see (9)–(11)] are used in this controller.

E. Regulation Service Market MPC

Regulation service helps to match generation and load and adjusts generation output to maintain the desired frequency. There is a penalty deviation cost used as an incentive for the market participants to maintain their power balance. A consideration must be taken into account of the importance of the **reference** levels marked for each ESS in every hour by the previous MPC controllers. MPC is executed every 10 min with a prediction horizon of 3 h discretized in periods of 10 min to track the reference set by the previous MPC using receding horizon. There is a reserved hydrogen storage band of 1-Nm³ amplification, which is not used in the rest of the controllers in order to manage bad forecast scenarios. Thus,

$$J(h_i, m_j) = J_{\text{grid}}^{\text{RM}}(h_i, m_j) + J_{\text{uc}}^{\text{RM}}(h_i, m_j) + J_{\text{bat}}^{\text{RM}}(h_i, m_j) + J_{H_2}^{\text{RM}}(h_i, m_j). \quad (24)$$

In the next subsections, (24) will be particularized for each ESS.

1) Grid Cost Function: The cost function of the grid is given by

$$J_{\text{grid}}^{\text{RM}}(t_k) = \Gamma_{\text{up,reg}}^{\text{RM}}(t_k) \cdot (P_{\text{grid}}(t_k) - P_{\text{grid}}^{\text{sch}}(t_k)) \cdot \delta_{\text{up,reg}}(t_k) - \Gamma_{\text{down,reg}}^{\text{RM}}(t_k) \cdot (P_{\text{grid}}(t_k) - P_{\text{grid}}^{\text{sch}}(t_k)) \cdot \delta_{\text{down,reg}}(t_k) \quad (25)$$

where

$$\delta_{\text{up,reg}}(h_i, m_j) = \begin{cases} 1, & (P_{\text{grid}}(h_i, m_j) - P_{\text{grid}}^{\text{sch}}(h_i)) \geq 0 \\ 0, & (P_{\text{grid}}(h_i, m_j) - P_{\text{grid}}^{\text{sch}}(h_i)) < 0 \end{cases} \quad (26)$$

$$\delta_{\text{down,reg}}(h_i, m_j) = \begin{cases} 0, & (P_{\text{grid}}(h_i, m_j) - P_{\text{grid}}^{\text{sch}}(h_i)) \geq 0 \\ 1, & (P_{\text{grid}}(h_i, m_j) - P_{\text{grid}}^{\text{sch}}(h_i)) < 0. \end{cases} \quad (27)$$

This expression gives results in the introduction of constraints (62)–(65) in the MPC controller.

2) Ultracapacitor Cost Function: The ultracapacitor cost function is included to maintain the ultracapacitor always available in order to balance the energy surplus or deficit between the long term and the short term, keeping the ultracapacitor into an intermediate SOC, i.e.,

$$J_{\text{uc}}(h_i, m_j) = \sum_{n=1}^{\text{SH}} \left(w_{\text{uc}} \cdot (\text{SOC}_{\text{uc}}(h_i, m_{j+10n}) - \text{SOC}_{\text{uc}}^{\text{ref}})^2 \right). \quad (28)$$

3) Batteries Cost Function: The functional cost of the regulation service made by the battery is the deviation at the end of the short-term horizon prediction (1 h) discretized in six periods of 10 min in the short-term MPC. Note that this deviation is taken only at the end of the control horizon. The aspects concerning the batteries degradation must be also considered by the controller. Thus,

$$\begin{aligned} J_{\text{bat}}(h_i, m_j) &= w_{\text{bat}} \cdot (\text{SOC}_{\text{bat}}(h_i, m_{j+60n}) - \text{SOC}_{\text{bat}}^{\text{IM}}(h_i, m_{j+60n}))^2 \\ &+ \frac{1}{6} \sum_{n=1}^{n=6} \left(\frac{\text{CC}_{\text{bat}}}{2 \cdot \text{Cycles}_{\text{bat}}} \times \left(\frac{P_{\text{bat,dis}}(h_i, m_{j+10n})}{\eta_{\text{dis,bat}}} \right. \right. \\ &\quad \left. \left. + P_{\text{bat,ch}}(h_i, m_{j+10n}) \cdot \eta_{\text{bat,ch}} \right) \right. \\ &\quad \left. + \text{Cost}_{\text{degr,dis}} \cdot P_{\text{bat,dis}}^2(h_i, m_{j+10n}) \right. \\ &\quad \left. + \text{Cost}_{\text{degr,ch}} \cdot P_{\text{bat,ch}}^2(h_i, m_{j+10n}) \right). \quad (29) \end{aligned}$$

4) Hydrogen Cost Function: The functional cost of the use of hydrogen ESS for the regulation market level control is a penalty from the final instant of the control horizon combined with the aspects of degradation and useful cost of this system, i.e.,

$$\begin{aligned} J_{H_2}(h_i, m_j) &= w_{H_2} \cdot (\text{LOH}(h_i, m_{j+60n}) - \text{LOH}^{\text{MM}}(h_i, m_{j+60n}))^2 \\ &+ \frac{1}{6} \sum_{n=1}^{n=6} \left(\left(\frac{\text{CC}_{\text{elz}}}{\text{Hours}_{\text{elz}}} + \text{Cost}_{\text{o\&m,elz}} \right) \delta_{\text{elz}}(h_i, m_{j+10n}) \right. \\ &\quad + \text{Cost}_{\text{startup,elz}} \cdot \sigma_{\text{elz}}^{\text{on}}(h_i, m_{j+10n}) \\ &\quad + \text{Cost}_{\text{shutdown,elz}} \cdot \sigma_{\text{elz}}^{\text{off}}(h_i, m_{j+10n}) \\ &\quad + \text{Cost}_{\text{degr,elz}} \cdot \vartheta_{\text{elz}}^2(h_i, m_{j+10n}) \\ &\quad \times \left(\frac{\text{CC}_{\text{fc}}}{\text{Hours}_{\text{fc}}} + \text{Cost}_{\text{o\&m,fc}} \right) \delta_{\text{fc}}(h_i, m_{j+10n}) \\ &\quad + \text{Cost}_{\text{startup,fc}} \cdot \sigma_{\text{fc}}^{\text{on}}(h_i, m_{j+10n}) \\ &\quad + \text{Cost}_{\text{shutdown,fc}} \cdot \sigma_{\text{fc}}^{\text{off}}(h_i, m_{j+10n}) \\ &\quad \left. + \text{Cost}_{\text{degr,fc}} \cdot \vartheta_{\text{fc}}^2(h_i, m_{j+10n}) \right). \quad (30) \end{aligned}$$

F. System Constraints

1) Physical Constraints: Physical constraints are given by the upper and lower limits of the ESS, which can absorb or supply the maximum and minimum levels of energy storage

that every ESS has got (main grid power exchange is also considered), i.e.,

$$P_{pv}(t_k) + P_{wt}(t_k) - P_{load}(t_k) = P_{grid}(t_k) + z_{elz}(t_k) - z_{fc}(t_k) + P_{bat}(t_k) + P_{uc}(t_k) \quad (31)$$

$$P_i^{\min} \leq P_i(t_k) \leq P_i^{\max} |_{i=grid,elz,fc,bat,uc} \quad (32)$$

$$SOC_i^{\min} \leq SOC_i(t_k) \leq SOC_i^{\max} |_{i=bat,uc} \quad (33)$$

$$LOH^{\min} \leq LOH(t_k) \leq LOH^{\max} \quad (34)$$

$$0 \leq \delta_i(t_k) \leq 1 |_{i=elz,fc} \quad (35)$$

2) MLD Constraints: The conversions introduced by Bemporad and Morari [54] make it possible to include binary and auxiliary variables introduced in a discrete-time dynamic system in order to describe, in a unified model, the evolution of the continuous and logic signals of the system. Thus,

$$0 \leq \delta_{sale}(t_k) + \delta_{pur}(t_k) \leq 1 \quad (36)$$

$$P_{sale}(t_k) - P_{pur}(t_k) = P_{grid}(t_k) \quad (37)$$

$$P_{grid}^{\min} \delta_{sale}(t_k) \leq P_{sale}(t_k) \leq P_{grid}^{\max} \delta_{sale}(t_k) \quad (38)$$

$$P_{grid}(t_k) - P_{grid}^{\max} (1 - \delta_{sale}(t_k)) \leq P_{sale}(t_k) \quad (39)$$

$$P_{sale}(t_k) \leq P_{grid}(t_k) - P_{grid}^{\min} (1 - \delta_{sale}(t_k)) \quad (40)$$

$$0 \leq \delta_{ch,i}(t_k) + \delta_{dis,i}(t_k) \leq 1 |_{i=uc,bat} \quad (41)$$

$$P_{ch,i}(t_k) - P_{dis,i}(t_k) = P_i(t_k) |_{i=uc,bat} \quad (42)$$

$$P_i^{\min} \delta_{ch,i}(t_k) \leq P_{ch,i}(t_k) \leq P_i^{\max} \delta_{ch,i}(t_k) |_{i=uc,bat} \quad (43)$$

$$P_i(t_k) - P_i^{\max} (1 - \delta_{ch,i}(t_k)) \leq P_{ch,i}(t_k) |_{i=uc,bat} \quad (44)$$

$$P_{ch,i}(t_k) \leq P_i(t_k) - P_i^{\min} (1 - \delta_{ch,i}(t_k)) |_{i=uc,bat} \quad (45)$$

$$0 \leq \delta_{elz}(t_k) + \delta_{fc}(t_k) \leq 1 \quad (46)$$

$$-\delta_i(t_k) + \sigma_i^{on}(t_k) \leq 0 |_{i=elz,fc} \quad (47)$$

$$-(1 - \delta_i(t_k - 1)) + \sigma_i^{on}(t_k) \leq 0 |_{i=elz,fc} \quad (48)$$

$$\delta_i(t_k) + (1 - \delta_i(t_k - 1)) - \sigma_i^{on}(t_k) \leq 1 |_{i=elz,fc} \quad (49)$$

$$-\delta_i(t_k - 1) + \sigma_i^{off}(t_k) \leq 0 |_{i=elz,fc} \quad (50)$$

$$-(1 - \delta_i(t_k)) + \sigma_i^{off}(t_k) \leq 0 |_{i=elz,fc} \quad (51)$$

$$\delta_i(t_k - 1) + (1 - \delta_i(t_k)) - \sigma_i^{off}(t_k) \leq 1 |_{i=elz,fc} \quad (52)$$

$$P_i^{\min} \delta_i(t_k) \leq z_i(t_k) \leq P_i^{\max} \delta_i(t_k) |_{i=pur,sale,elz,fc} \quad (53)$$

$$P_i(t_k) - P_i^{\max} (1 - \delta_i(t_k)) \leq z_i(t_k) |_{i=pur,sale,elz,fc} \quad (54)$$

$$z_i(t_k) \leq P_i(t_k) - P_i^{\min} (1 - \delta_i(t_k)) |_{i=pur,sale,elz,fc} \quad (55)$$

$$-\delta_i(t_k) + \chi_i(t_k) \leq 0 |_{i=elz,fc} \quad (56)$$

$$-\delta_i(t_{k-1}) + \chi_i(t_k) \leq 0 |_{i=elz,fc} \quad (57)$$

$$\delta_i(t_k) + \delta_i(t_{k-1}) - \chi_i(t_k) \leq 1 |_{i=elz,fc} \quad (58)$$

$$\Delta z_i^{\min}(\chi_i(t_k)) \leq \vartheta_i(t_k) \leq \Delta z_i^{\max}(\chi_i(t_k)) |_{i=elz,fc} \quad (59)$$

$$\Delta z_i(t_k) - \Delta z_i^{\max} (1 - \chi_i(t_k)) \leq \vartheta_i(t_k) |_{i=elz,fc} \quad (60)$$

$$\vartheta_i(t_k) \leq \Delta z_i(t_k) - \Delta z_i^{\min} (1 - \chi_i(t_k)) |_{i=elz,fc} \quad (61)$$

$$(P_{grid}(h_i, m_j) - P_{grid}^{sch}(h_i)) \leq P_{grid}^{\max} - P_{grid}^{\max} \cdot \delta_{down}(t_k) \quad (62)$$

$$(P_{grid}(h_i, m_j) - P_{grid}^{sch}(h_i)) \geq \epsilon + (-\epsilon) \cdot \delta_{down}(t_k) \quad (63)$$

$$-(P_{grid}(h_i, m_j) - P_{grid}^{sch}(h_i)) \leq P_{grid}^{\max} - P_{grid}^{\max} \cdot \delta_{up}(t_k) \quad (64)$$

$$-(P_{grid}(h_i, m_j) - P_{grid}^{sch}(h_i)) \geq \epsilon + (-\epsilon) \cdot \delta_{up}(t_k) \quad (65)$$

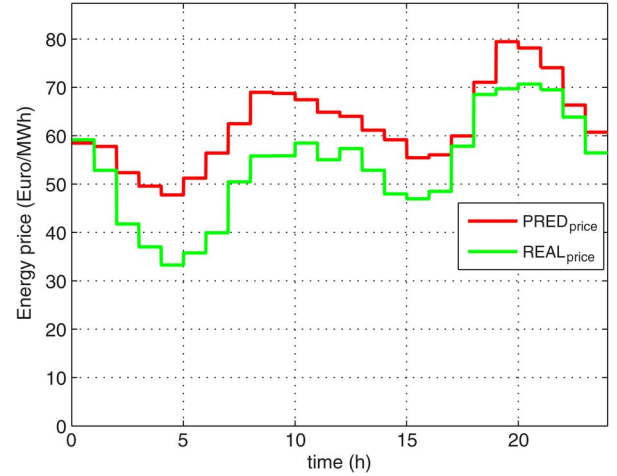


Fig. 3. Daily market ARIMA model price forecast.

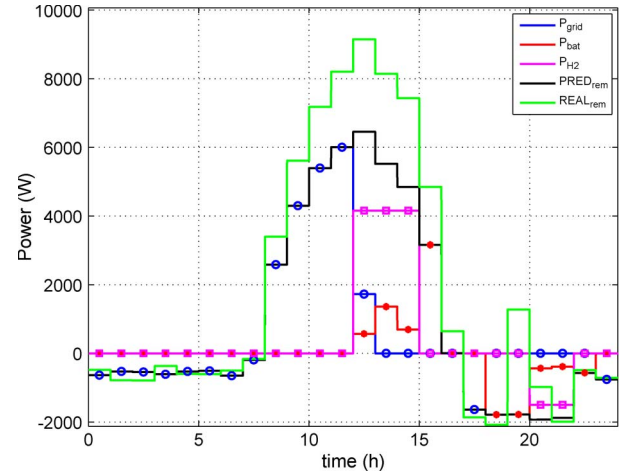


Fig. 4. Daily market MPC controller schedule.

IV. MPC SCHEDULE SIMULATION RESULTS

A. Daily Market MPC Results

The results of the ARIMA model and the daily market MPC controller are shown in Figs. 3 and 4. If the scenario is a surplus of energy, the system tries to sell energy to the grid, but when there is a deficit of energy in the microgrid, the system tries to use the hybrid ESS. The number of hours of use and the switching states of the electrolyzer and the fuel cell are minimized, as well as the peak power in the charge and discharge of the battery. The SOC of the batteries is also controlled imposing constraints to the controller protecting them from high states of charge or discharge. As shown in Fig. 4, the purchase of energy to the grid is done when the prices are lower, in the same way that the sale of energy to the grid is given at the maximum price periods. The power reference given to the fuel cell and the electrolyzer is set constant, giving the minimum variation to these devices and minimizing in this way the degradation due to the operation of the electrolyzer and the fuel cell.

B. Intraday Market MPC Results

The ARIMA forecast model results are more accurate when the prediction horizon is lower. For the day selected, a bad

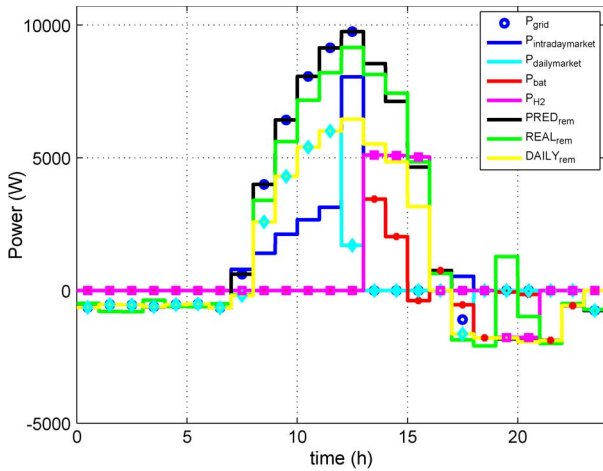


Fig. 5. Intraday market MPC controller schedule in surplus scenario.

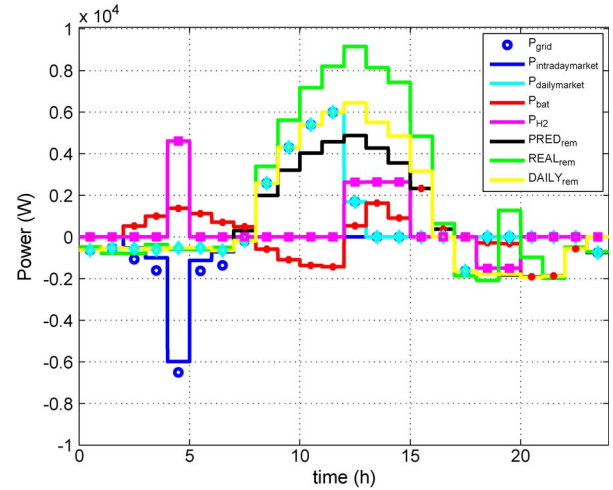


Fig. 7. Intraday market MPC controller schedule in deficit scenario.

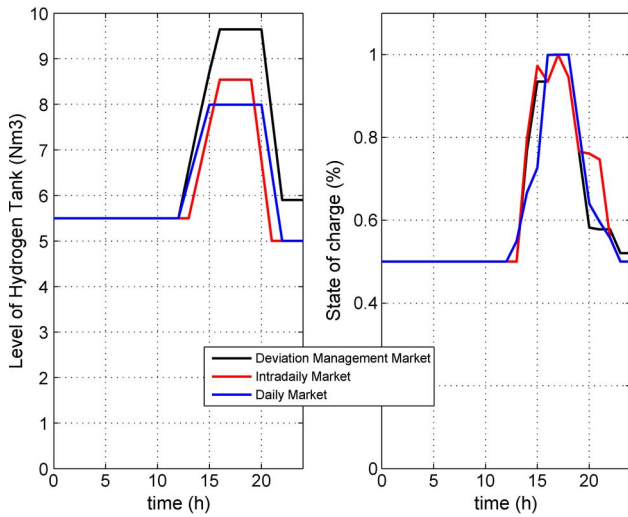


Fig. 6. Daily, intraday market, and deviation management MPC controller schedule level for LOH and SOC in surplus scenario.

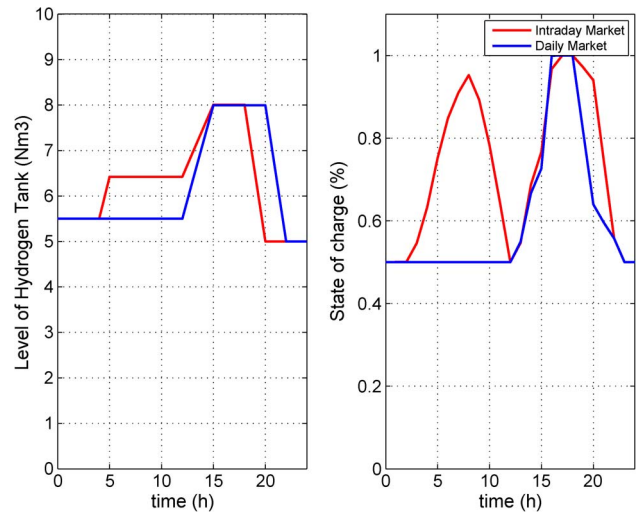


Fig. 8. Daily and intraday market MPC controller schedule level for LOH and SOC in deficit scenario.

forecast scenario is given for the daily market session (38 h ahead), but this is corrected in the medium-term horizon (28 h ahead) given in the first session of the intraday market. A new schedule is given for the first session of the intraday market, as shown in Fig. 5, where the price forecast is based on the results of the previous session of the market. Respecting the degradation issues and the different constraints of each ESS, the economical benefit of the microgrid is given, shifting the hours of the energy storage to that where the prices are lower from the price forecast scenario of the daily market. Although the horizon prediction of the remaining power would be shorter, the renewable energy prediction and the load forecast are always subject to errors. The rest of the control levels of the microgrid must manage these kinds of events in order not to generate economical losses in the microgrid at the time that the degradation issues for the different ESS are considered.

Two cases are shown for the intraday market algorithm results: the surplus scenario and the deficit scenario (see Figs. 5 and 7), with the evolution of the state variables for both cases (see Figs. 6 and 8). The total energy exchange with

the main grid is given adding the results of the daily market (cyan) and the intradaily market (blue). Note that, for the case of surplus, the maximum price value for the intraday market is different for the daily market; thus, the maximum power sale of the intraday market is carried at this moment. A reschedule of the energy stored is done according to the new prediction data with repercussion with the main grid instants. Note that, for the case of the deficit scenario (see Figs. 7 and 8), energy must be purchased in the intradaily market to fulfill the schedule done in the daily market. In order to minimize the losses, electricity is bought during the night when the prices are lower to charge the batteries and the hydrogen deposit. In the case of the batteries, this energy is used to minimize the purchase made to fulfill the requirement of the daily market schedule.

C. Deviation Management Market MPC Results

The results of the deviation management market control level are given in Fig. 9. In this case, the deviation management

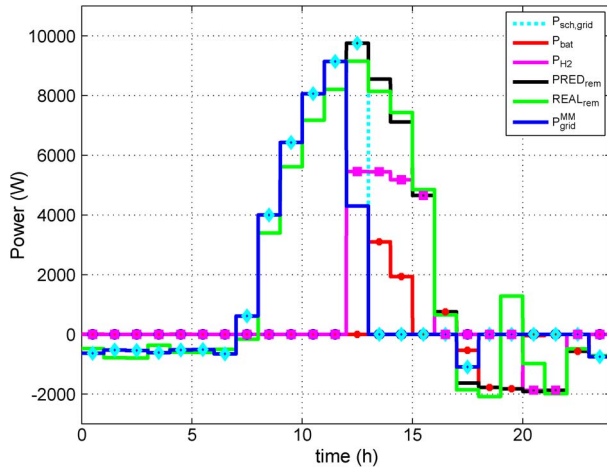


Fig. 9. Deviation management market.

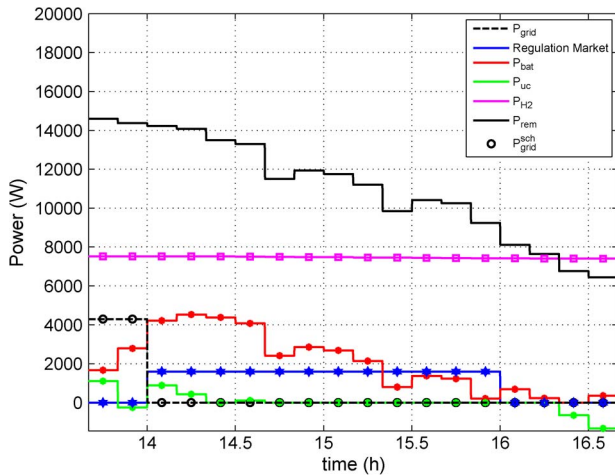


Fig. 10. Regulation service market MPC controller schedule level.

market is launched between 9 and 14 h after the fourth session of the intraday market. The results of the algorithm give a deviation down at 13.00 h, where the scheduled energy to be sold to the grid is transformed into hydrogen by the electrolyzer, as shown in Fig. 6.

D. Regulation Service MPC Results

The regulation service MPC is the last schedule control level before the real-time optimal load sharing. It is launched half an hour before each scheduled hour with a control horizon of 3 h in order to ensure that the schedule is going to be reached. The short-term horizon of this controller includes the action of the ultracapacitor ESS. As shown in Fig. 10, the algorithm gives a constant reference for the electrolyzer for all the working hours, minimizing the degradation effects. The downregulation with the main grid is minimized, whereas the upregulation is maximized and kept constant during 2 h. The peak current charge of the batteries is also minimized, giving a smooth current profile to this kind of ESS. The ultracapacitor ESS is maintained over the reference SOC, and the level of stored energy in batteries and hydrogen is followed (see Fig. 11).

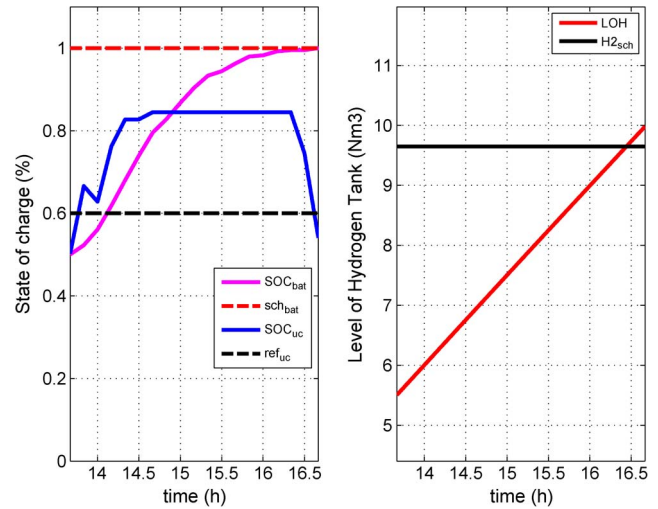


Fig. 11. Evolution of the state variables in the regulation service market MPC controller schedule level.

V. COMPARISON WITH OTHER CONTROL METHODS

Every element in the microgrid has its own casuistry, constraint, and cost. That imposes on the controller multiple variables to optimize and multiple constraints to fulfill. Several studies associated to management of microgrids with hybrid storage propose the hysteresis method (see [26]–[28]), where the hydrogen ESS is activated depending on the SOC of the batteries, and the batteries are activated depending on the SOC of the ultracapacitor. However, this method does not take into account the associated cost of the use of each ESS. The use of heuristical methods to solve a similar objective algorithm would have too many subcases, which increases its complexity and poses difficulty in guaranteeing reaching the optimal schedule. This complexity is increased with the use of a high number of constraints. The use of these kinds of algorithms is appropriate when there is not a wide number of variables to optimize and constraints to fulfill, but when the complexity grade of the control algorithm is elevated without a guarantee of reaching the optimal solution establishing local minimum. The mathematical formulation used by MPC allows minimizing a cost function with an optimization solver guaranteeing that the optimal solution is found even in the presence of the MLD framework. The main disadvantages of the use of MPC controllers are the formulation complexity and the fact that an optimization solver package is necessary. None of the work presented in the literature has considered the degradation issues incorporated into the cost function.

In order to quantitatively evaluate the performance of the proposed controller, a comparison for the daily market schedule with three control strategies is carried out. The first strategy is based on the manual case where a fixed hybridization (FH) degree between the battery and the fuel cell is imposed. This strategy sells to the grid when the price prediction is higher than a price reference and buys from the grid when this prediction is lower than a fixed reference. The second strategy is based on the hysteresis method using also the price hysteresis band. The third strategy is based on MILP (see [43]–[46]), where only

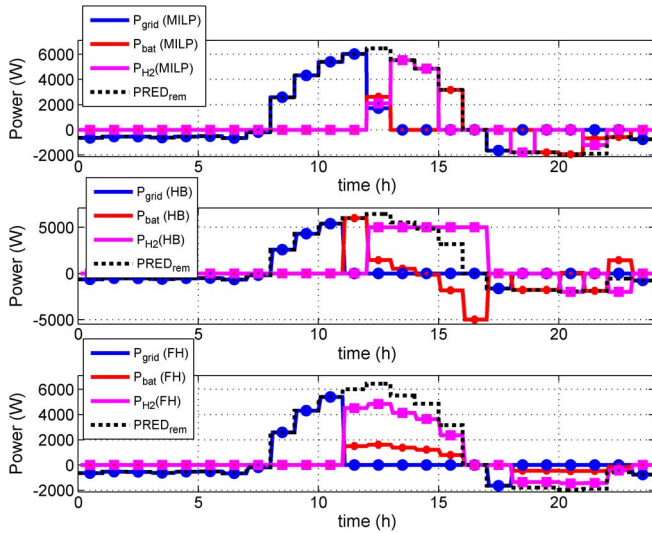


Fig. 12. Daily market schedule results for different controllers.

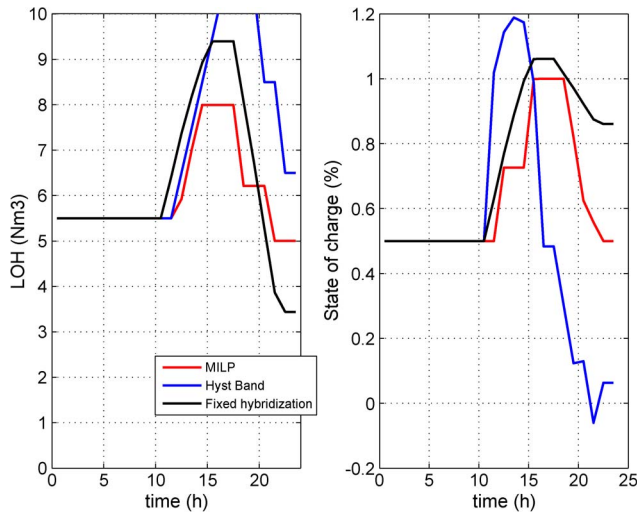


Fig. 13. Schedule level for LOH and SOC for different controllers.

operational costs without degradation issues are introduced. The results of these three strategies can be observed in Figs. 12 and 13, which can be compared with the results obtained for the **mixed-integer quadratic programming (MIQP)** proposed controller in Figs. 4 and 6. In Table II, several key performance indicators are exposed for each one of the strategies under study. Similar efficiency is found for the MILP and the MIQP, but a better distribution of the ESS for minimum degradation is found in the MIQP. The MIQP controller proposed in this paper is the only one of the four proposed strategies that is always the most favorable (8/11) or the second most favorable (3/11) controller for all the key indicators studied.

VI. CONCLUSION

In this paper, an MPC controller for the optimal economical schedule of a microgrid has been presented **and validated**. This

TABLE II
COMPARISON OF THE KEY PERFORMANCE
VALUES OF THE CONTROLLERS

	MIQP	MILP	HB	FH
Economic Balance Grid(€)	0.9685*	0.9685*	0.4686	0.4686
Hours of the ELZ	3*	3*	5**	5**
Start up/shut down ELZ	1*	1*	1*	1*
Reference changes ELZ	2*	4**	2*	6
Hours of the FC	2*	2*	2*	4**
Start up/shut down FC	1*	2**	2**	1*
Reference changes FC	2*	4**	4**	6
Charge cycle(%)	50*	50*	56.12**	50*
Discharge cycle(%)	50**	50**	100	16.25*
Maximum charge BAT	3160**	3160**	6004	1613*
Maximum discharge BAT	1784**	1931	4994	482.75*

(*) Most favourable controller

(**) Second most favourable

formulation integrates the operational costs in the optimization problem, including degradation of all the components of the hybrid ESS, particularly the hydrogen-based storage. This allows an optimal scheduling policy linked to the time-varying price of energy. The results show an optimal behavior of the microgrid whose nondispatchable generation is converted into dispatchable using the ESS.

The results obtained of the function cost minimization conclude that hydrogen ESS is appropriate for great quantities of energy storage requirements in short functional periods, whereas the batteries are appropriate for low quantities in the energy storage requirements with long functional periods. **Ultracapacitors are the most appropriate technology to correct rapid fluctuance in the disposal of energy in the microgrid. The complementarity between the three ways to store the energy is also manifested in this paper. When the hydrogen storage is active, the overcharge and the undercharge of the battery and the ultracapacitor can be controlled. The high stress current ratio in the battery is also neglected with the use of the hydrogen ESS. The fluctuance operation conditions in the hydrogen are avoided with the use of batteries and ultracapacitor.**

According to the economical values used in this paper to make the ESS competitive, their lifetime must be increased, decreasing in this way their **associated** cost. This paper has shown how the control algorithm can contribute to this objective.

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